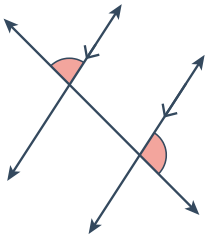


Parallel lines and transversals

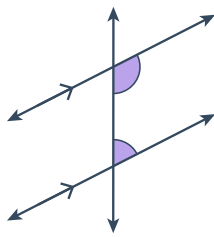


1 Identify the types of angles formed by the transversal in the following diagrams:

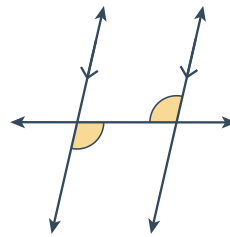
a



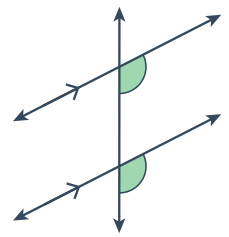
b



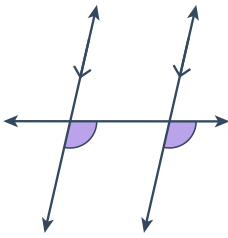
c



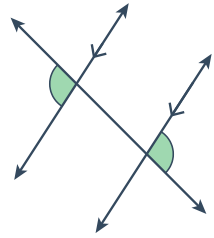
d



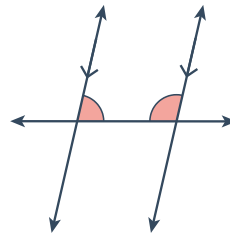
e



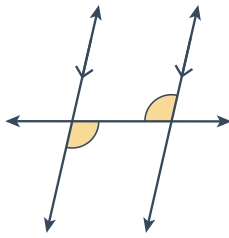
f



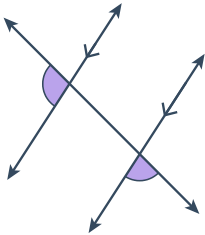
g



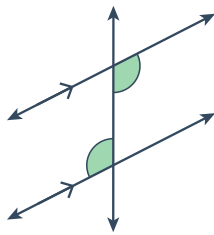
h



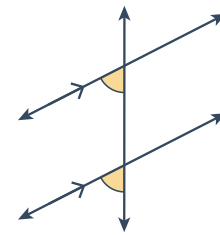
i



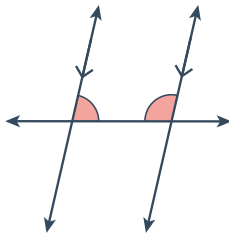
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k



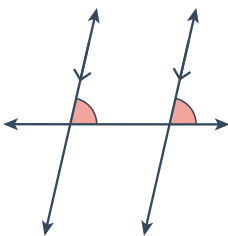
l



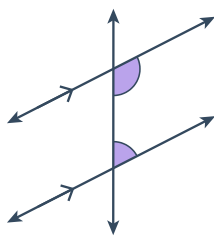
2 Consider the following diagrams:

- State whether the two highlighted angles are congruent, complementary or supplementary.
- Identify the types of angles formed by the transversal.

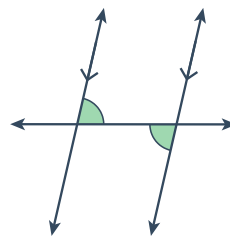
a



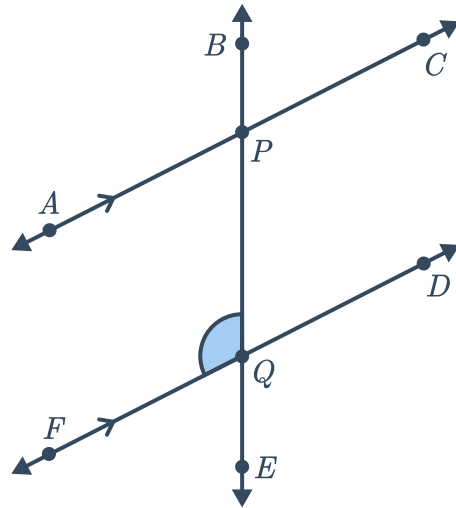
b



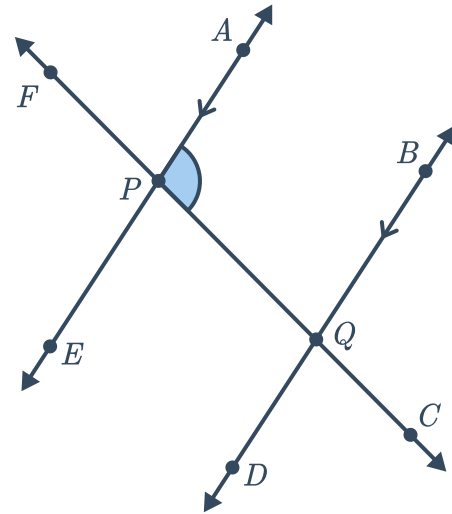
c



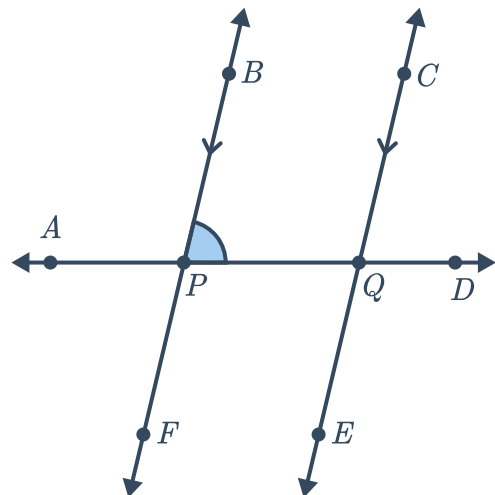
- 3 Consider the given diagram:
Write the angle that is corresponding to $\angle BQF$.



- 4 Consider the given diagram:
Write the angle that is alternate interior to $\angle APC$.



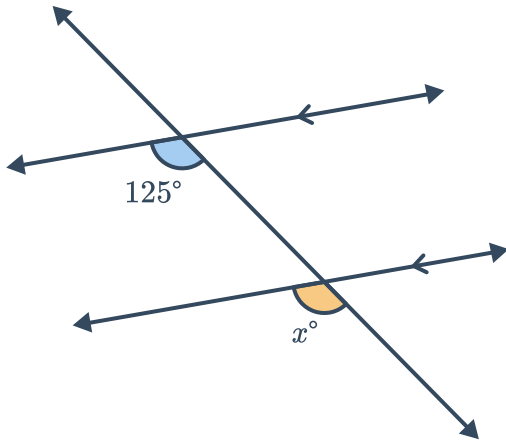
- 5 Consider the given diagram:
Write the angle that is consecutive interior to $\angle BPD$.



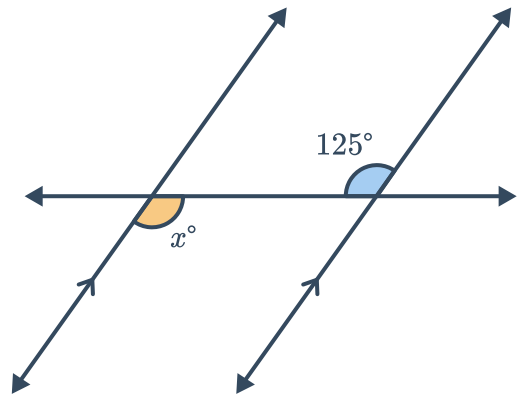
6 For the following, solve for the value of x .



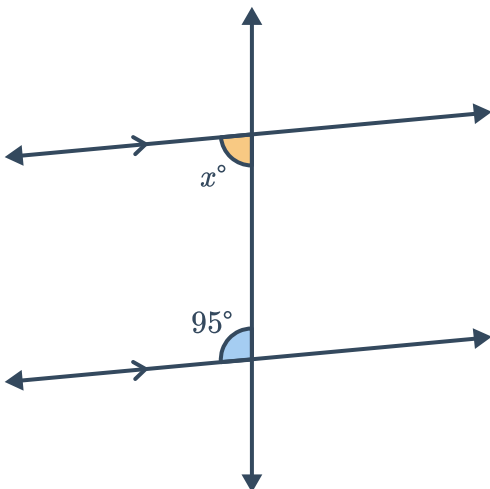
a



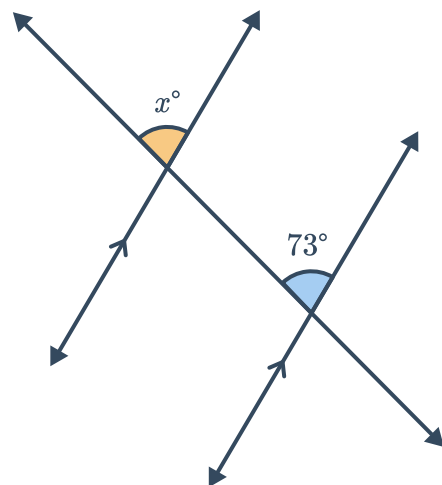
b



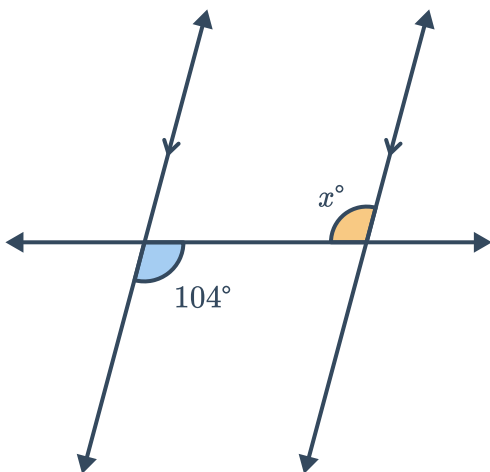
c



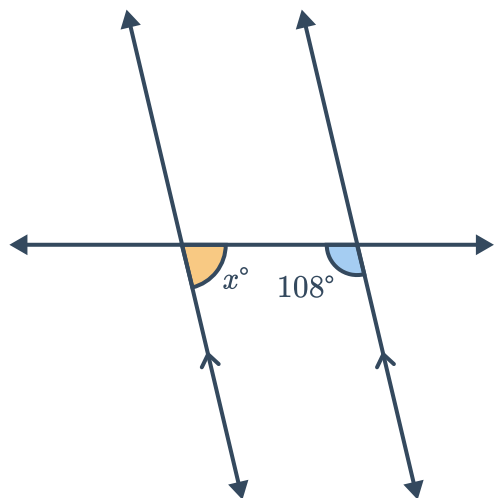
d



e



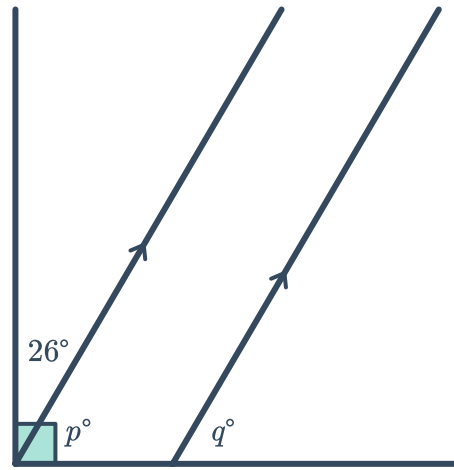
f



7 Consider the given diagram:

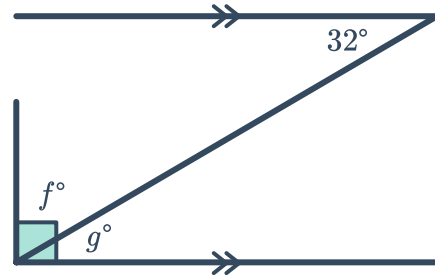


- a Solve for p .
- b Solve for q .



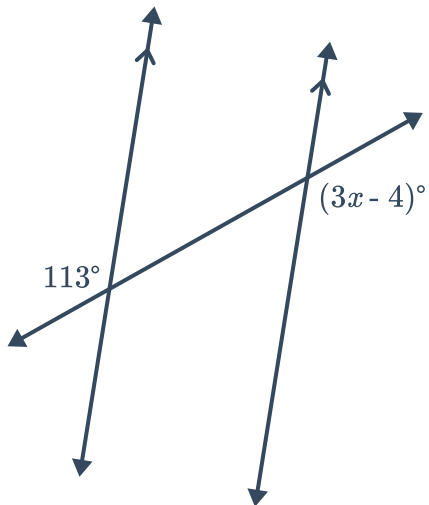
8 Consider the given diagram:

- a Solve for g .
- b Solve for f .

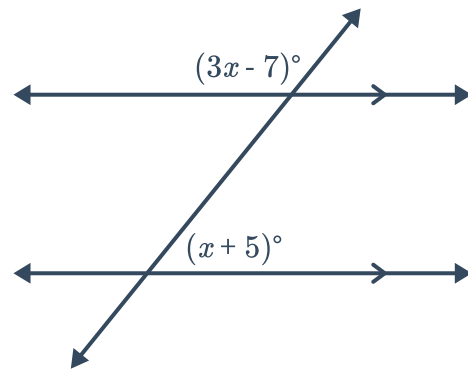


9 For the following, solve for x .

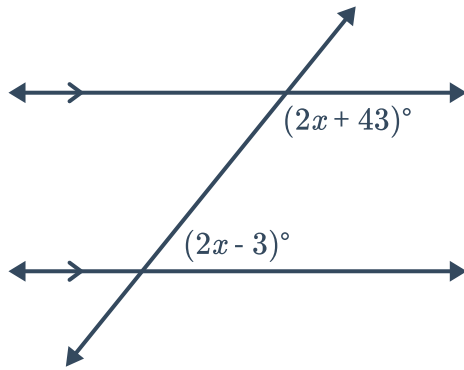
a



b

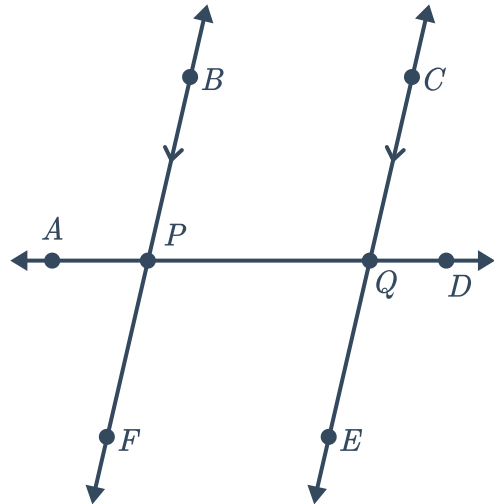


c



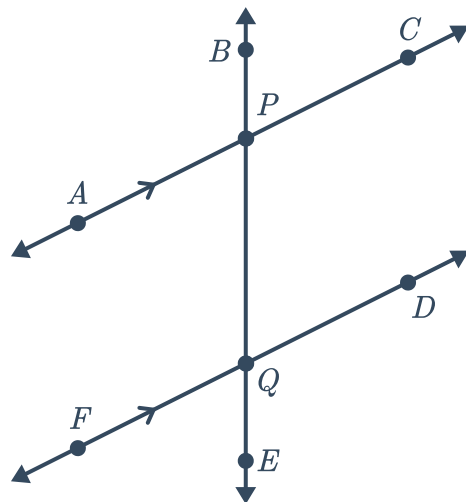
10 Consider the given diagram:

- a How are $\angle DPF$ and $\angle AQC$ related?
- b How are $\angle AQC$ and $\angle CQD$ related?
- c If $\angle DPF = 102^\circ$, find $m\angle CQD$.



11 Consider the given diagram:

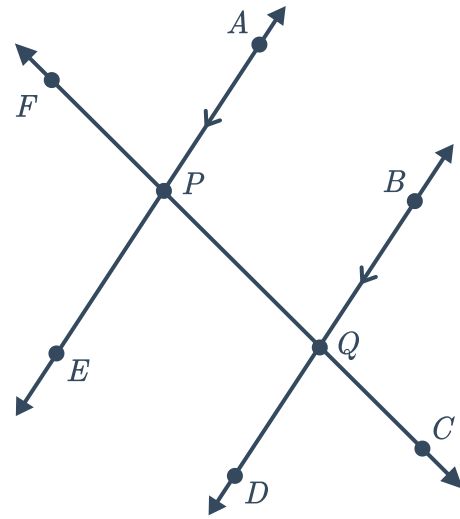
- a How are $\angle APB$ and $\angle CPE$ related?
- b How are $\angle CPE$ and $\angle BQD$ related?
- c If $m\angle APB = 106^\circ$, find $m\angle BQD$.



12 Consider the given diagram:

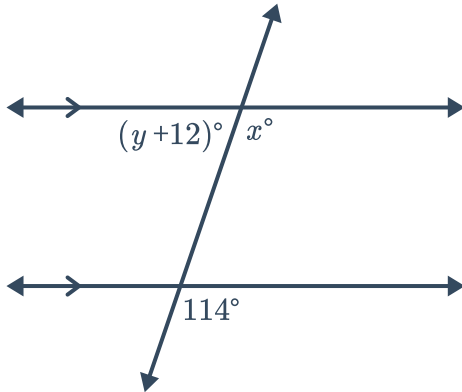


- a How are $\angle DQF$ and $\angle CPE$ related?
- b How are $\angle CPE$ and $\angle EPF$ related?
- c If $m\angle DQF = 102^\circ$, find $m\angle EPF$.

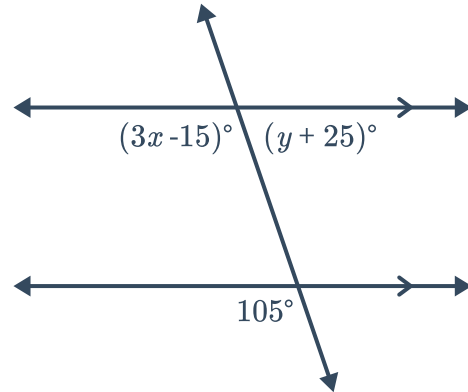


13 For the following diagrams, solve for the value of x and y .

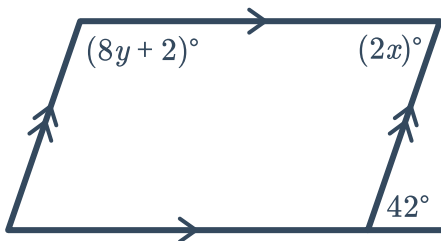
a



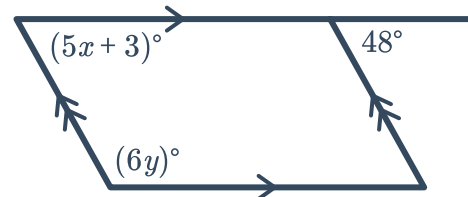
b



c



d

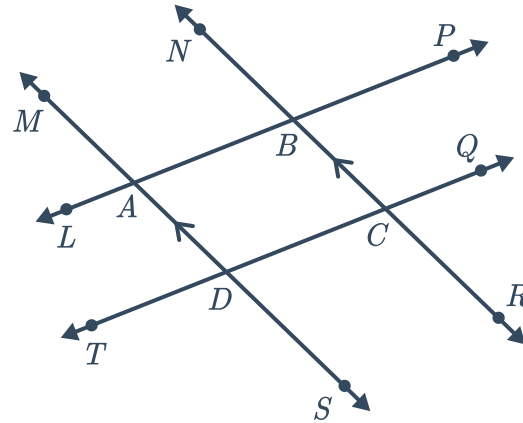


14 Consider the following diagram:



Given that $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$, which of the following are pairs of congruent angles?

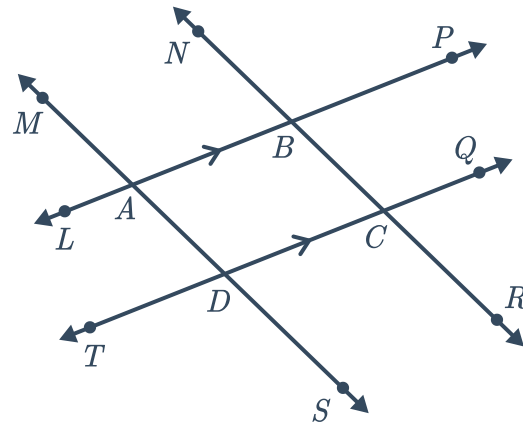
- $\angle ADC$ and $\angle DCR$
- $\angle NBA$ and $\angle QCR$
- $\angle BAD$ and $\angle ADT$
- $\angle ABC$ and $\angle BAD$



15 Consider the following diagram:

Given that $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$, which of the following are pairs of congruent angles?

- $\angle NBP$ and $\angle DCR$
- $\angle BCQ$ and $\angle TDS$
- $\angle NBA$ and $\angle BAD$
- $\angle MAL$ and $\angle TDS$



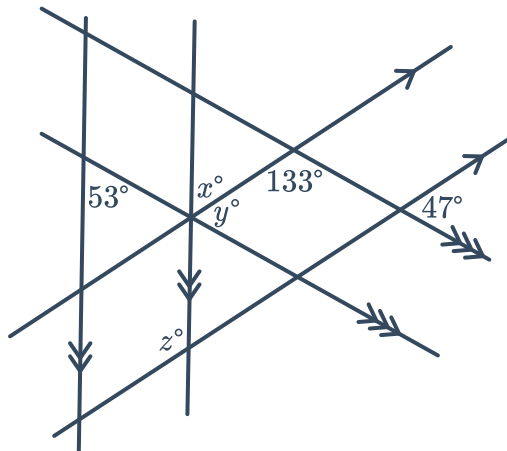
- 16 Consider the given diagram, which has three pairs of parallel lines. Find the value of the following:



a y

b x

c z

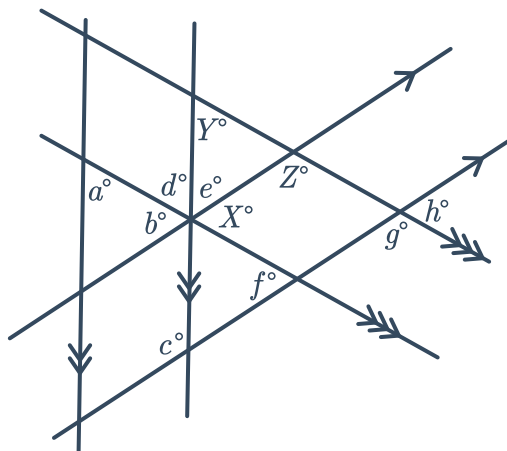


- 17 Consider the given diagram, which has three pairs of parallel lines. Write all variables that have the same value as the following:

a X

b Y

c Z



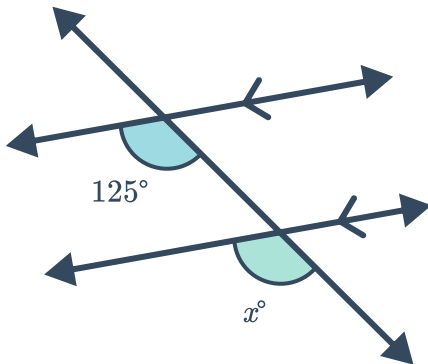


Additional questions

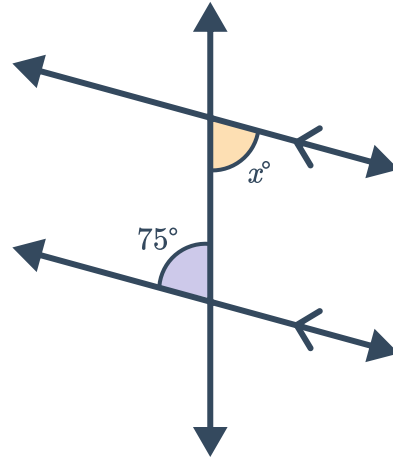
18 For each of the following diagrams:

- Identify the types of angles formed by the transversal in the following diagrams
- Describe how the highlighted angles are related to each other
- Find the value of x

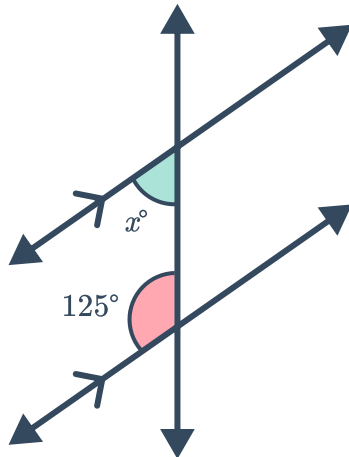
a



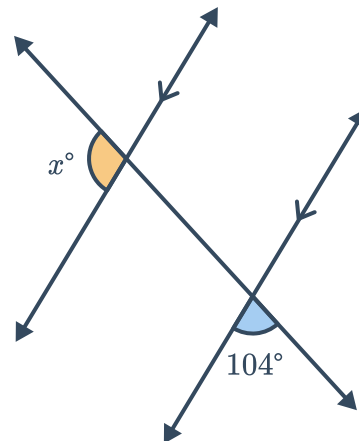
b



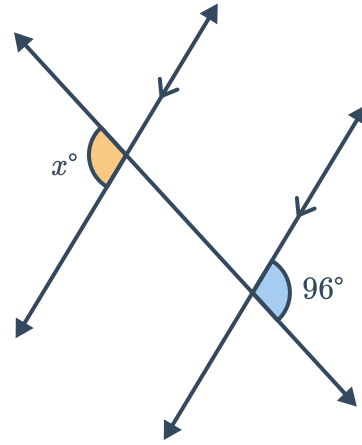
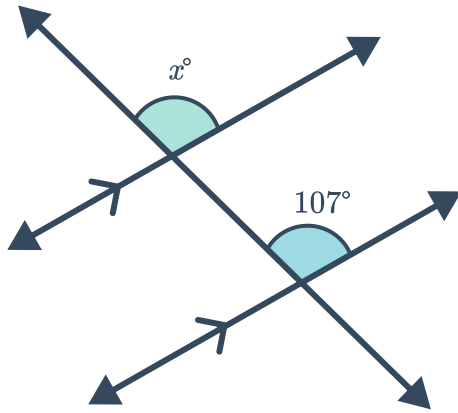
c



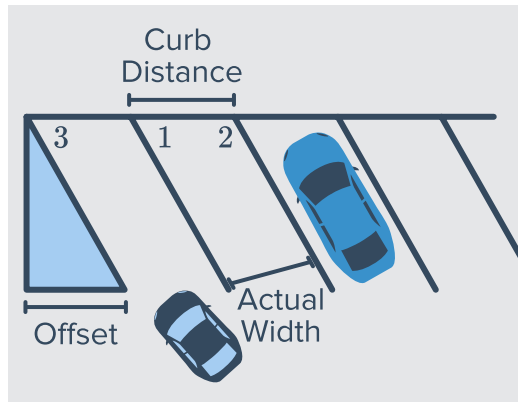
d



e

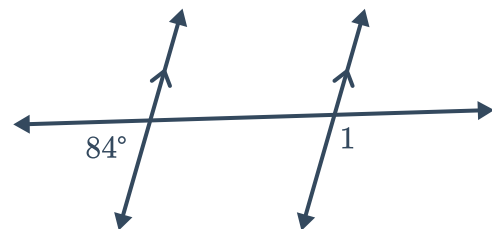


- 19 In the figure of the parking lot shown, the lines from the parking spaces should be parallel. If $m\angle 3 = 51^\circ$, what should $m\angle 1$ and $m\angle 2$ be? Explain your reasoning.

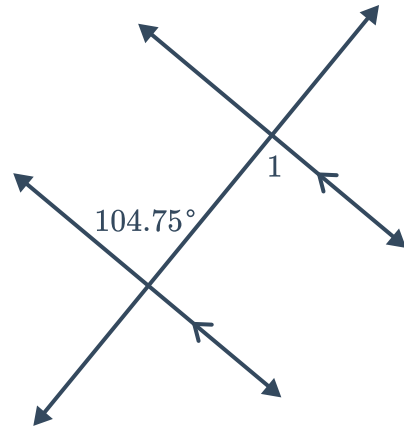


- 20 Review the following diagrams and statements. Determine whether the statements are correct. Explain your reasoning.

- a $m\angle 1 = 84^\circ$ by the consecutive exterior angles theorem

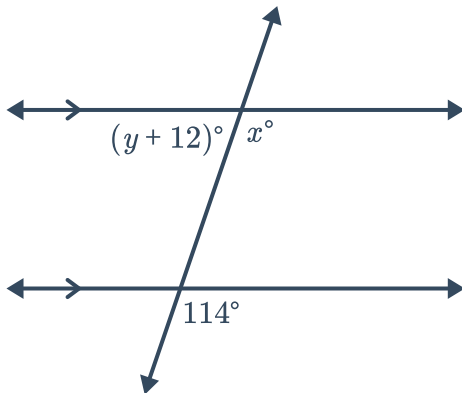


- b $m\angle 1 = 75.25^\circ$ by the alternate interior angles theorem

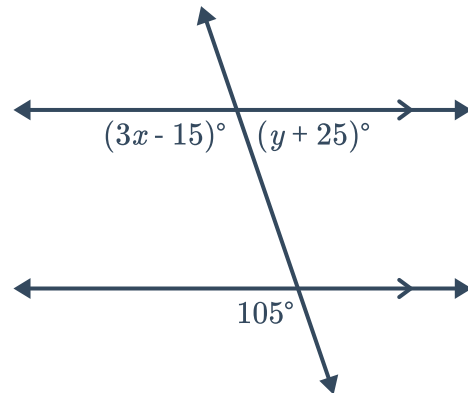


- 21 Consider the following diagrams:
Find the value of the unknown variable(s)

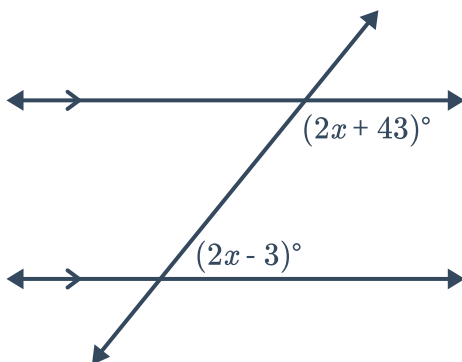
a



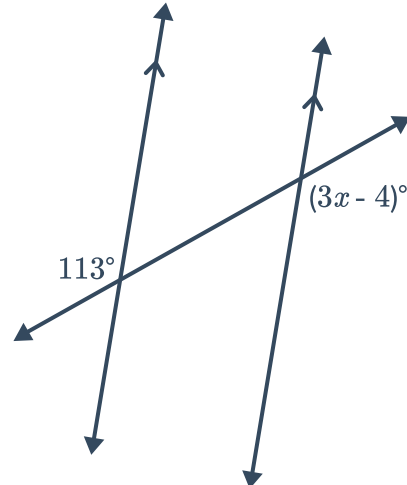
b



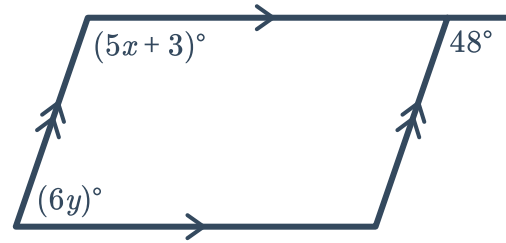
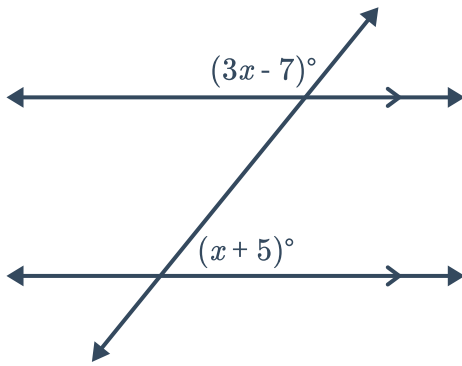
c



d

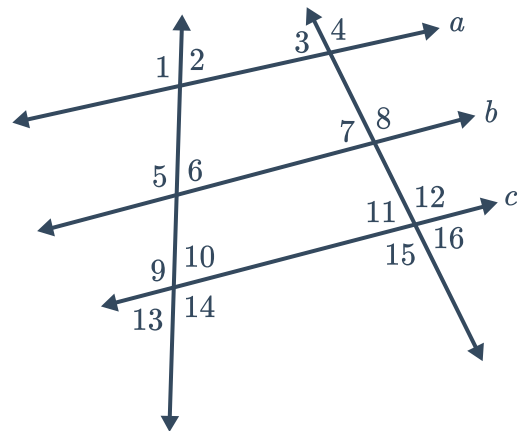


e



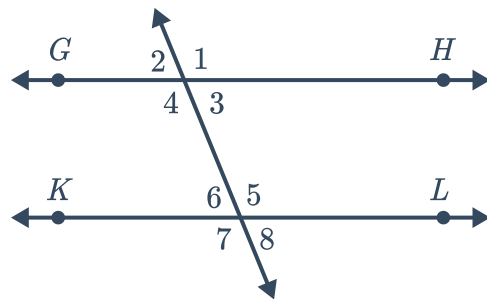
22 Determine if the information given is enough to justify the conclusion.

- a** Given: $a \parallel b$ and $\angle 1 \cong \angle 9$
Conclusion: $\angle 5 \cong \angle 9$
- b** Given: $\angle MNO$ and $\angle ONP$ are corresponding angles
Conclusion: $\angle MNO \cong \angle ONP$



23 Prove the alternate exterior angles theorem:

Given: $\overleftrightarrow{GH} \parallel \overleftrightarrow{KL}$
Prove: $\angle 2 \cong \angle 8$

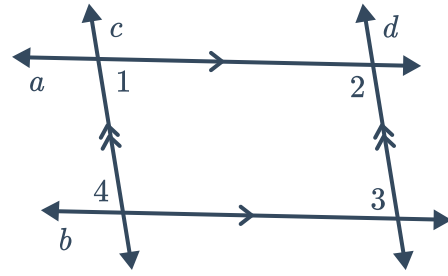


24

a Given: $a \parallel b, c \parallel d$

Prove: $\angle 1 \cong \angle 3$

b Prove part (a) in a different way.



25 Given: $\overleftrightarrow{AF} \parallel \overleftrightarrow{GL}$ and $\angle BHJ \cong \angle HJE$

Prove: $\angle HBE \cong \angle BEJ$

